Problem Set 3 Solutions

Due January 30, 2026 at 11:59 PM Pacific

Turn in your problem set by Gradescope. Include any code you write in your solutions as well. Mark down how much time you spent on the homework set.

1. [10 Points.] Consider the position of a mobile robot x and a GPS measurement of that position y. Suppose these two random vectors are jointly Gaussian with parameters

$$\mu_x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \Sigma_{xx} = \begin{bmatrix} 0.7 & 0.73 \\ 0.73 & 1.1 \end{bmatrix}, \quad \mu_y = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \Sigma_{yy} = \begin{bmatrix} 0.7 & 0.19 \\ 0.19 & 0.16 \end{bmatrix}, \quad \Sigma_{xy} = \begin{bmatrix} 0.63 & 0.23 \\ 0.72 & 0.31 \end{bmatrix}.$$

Given a GPS position measurement $y = [0.27 \quad 0.62]^T$, find the posterior mean and covariance $(\mu_{x|y}, \Sigma_{x|y})$ of the rover position. Plot the following:

- 95% error ellipse for the prior $\mathcal{N}(\mu_x, \Sigma_{xx})$
- 95% error ellipse for the posterior $\mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$
- the mean of the posterior $\mu_{x|y}$
- the GPS measurement y

Solution

Code solutions are attached. After running our script, we get

$$\mu_{x|y} = \begin{bmatrix} 0.5405 \\ 0.8554 \end{bmatrix}, \quad \Sigma_{x|y} = \begin{bmatrix} 0.1009 & 0.01966 \\ 0.01966 & 0.2384 \end{bmatrix}.$$

Plotting the prior and posterior error ellipses yields,

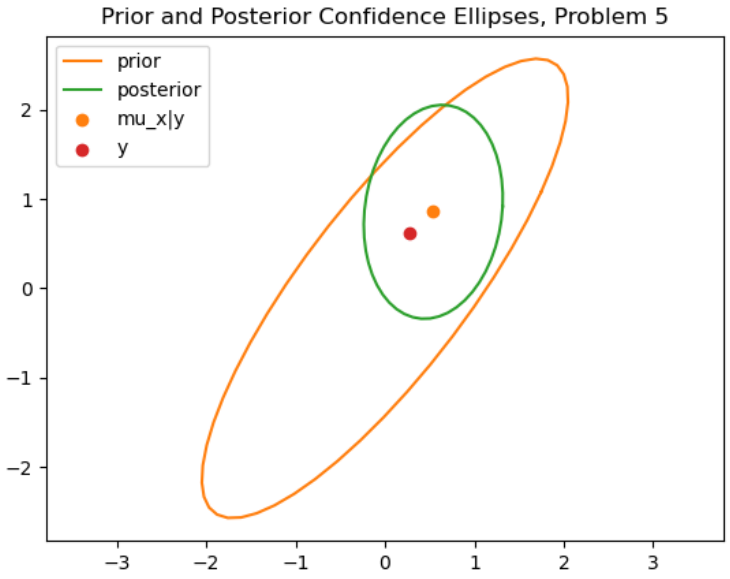


Figure 1: Prior and Posterior Gaussians, Pb. 5.

2. [10 Points.] In this problem, you will prove that $E[(X_t - \mu_{t|t})W_t^T \mid Y_{1:t}] = 0$, which will be useful in the derivation of the Kalman Filter in lecture.

   (a) [5 Points.] Show that for three (not necessarily Gaussian) random variables (X, Y, W), if $p(x, y, w) = p(x, y)p(w)$ then $\text{Cov}(X, W \mid Y) = 0$. Hint: write out the integral for $\text{Cov}(\cdot)$ and manipulate the distribution under the integral.

   (b) [5 Points.] Use the above result to prove $E[(X_t - \mu_{t|t})W_t^T \mid Y_{1:t}] = 0$.

   Solution

   (a) Given the assumption $p(x, y, w) = p(x, y)p(w)$, we have:

$$p(x, w|y) = \frac{p(x, y, w)}{p(y)} = \frac{p(x|y)p(y)p(w)}{p(y)} = p(x|y)p(w) \tag{1}$$

   Moreover, if we marginalize both sides over X, we can see that Y and W are independent:

$$\int_x p(x, y, w)dx = \int_x p(x, y)p(w)dx$$

$$p(y, w) = p(y)p(w) \tag{2}$$

   Thus, we have $\mu_{w|y} = \mu_w$(we will use this later).

$$\begin{aligned}
Cov(X, W|Y) =& E[(X - \mu_{x|y})(W - \mu_{w|y})^T|Y] \\
=& \int_x \int_w (x - \mu_{x|y})(w - \mu_w)^T p(x, w|y)dxdw \\
=& \int_x \int_w (xw^T - \mu_{x|y}w^T - x\mu_w^T + \mu_{x|y}\mu_w^T)p(x|y)p(w)dxdw \\
=& \int_x \int_w xw^T p(x|y)p(w)dxdw - \int_x \int_w \mu_{x|y}w^T p(x|y)p(w)dxdw \\
& - \int_x \int_w x\mu_w^T p(x|y)p(w)dxdw + \int_x \int_w \mu_{x|y}\mu_w^T p(x|y)p(w)dxdw \\
=& \int_x \int_w E[X|Y]w^T p(w)dxdw - \mu_{x|y} \int_x \int_w w^T p(x|y)p(w)dxdw - \\
& \int_x \int_w xp(x|y)p(w)dxdw\mu_w^T + \mu_{x|y}\mu_w^T \\
=& E[X|Y]\mu_w{}^T - \mu_{x|y}\mu_w{}^T - E[X|Y]\mu_w{}^T + \mu_{x|y}\mu_w^T \\
=& 0
\end{aligned}$$

   equation 3 holds true because of equation (1).

   (b) In Linear Gaussian Model, white noise is independent from all previous states and measurements. Thus, we have,

$$p(x_t, y_{1:t}, w_t) = p(w_t)p(x_t, y_{1:t})$$

   From (a), we proved that $Cov(X_t, W_t|Y_{1:t}) = 0$, and therefore we have $Cov(X_t, W_t|Y_{1:t}) = E[(X_t - \mu_{X_t|Y_{1:t}})(W_t - \mu_{W_t|Y_{1:t}})^T \mid Y_{1:t}] = E[(X_t - \mu_{t|t})(W_t - \mu_{W_t})^T \mid Y_{1:t}] = E[(X_t - \mu_{t|t})(W_t)^T \mid Y_{1:t}] - E[(X_t - \mu_{t|t}) \mid Y_{1:t}]\mu_{W_t}^T = 0$ (as shown before, $\mu_{w|y} = \mu_w$). By definition, $\mu_{t|t} = E[X_t|Y_{1:t}]$, so the second term is equal to 0, and we find that $E[(X_t - \mu_{t|t})(W_t)^T \mid Y_{1:t}] = 0$.

3. [10 Points.] Consider a quadrotor drone flying at a constant altitude. Let $p_t \in \mathbb{R}^2$ be its position, and $s_t \in \mathbb{R}^2$ its velocity. Let's model its dynamics as a discrete time double integrator perturbed by noise,

$$P_{t+1} = P_t + \delta t S_t \tag{3}$$

$$S_{t+1} = S_t + \delta t u_t + W_t, \tag{4}$$

where $u_t \in \mathbb{R}^2$ is the acceleration commanded by a remote human pilot, and δt is the discrete time step. The drone makes position measurements with GPS according to the noisy measurement model

$$Y_t = P_t + V_t. \tag{5}$$

The processes W_t, V_t are Gaussian white noise with covariances Q and R, respectively, uncorrelated with each other and uncorrelated with the initial state. Let $\delta t = 1s$, $Q = I_2 m^2/s^2$, $R = 9\mathrm{I}_2 m^2$, $u_t = -2.5[\cos(0.05t) \;\; \sin(0.05t)]^T m/s^2$.

(a) [5 Points.] Find the system matrices A, B, C, in terms of the given parameters.

(b) [5 Points.] Simulate the trajectory of the drone (in Python) as well as the measurement process. Let the initial position and velocity be $p_0 = [1000m \quad 0m]^T$, and $v_0 = [0m/s \quad 50m/s]^T$, respectively. Attach plots of a few example trajectories along with the noisy GPS position measurements.

Hint: The Python command `numpy.random.randn` will come in handy for producing the noise signals.

Solution

(a) Our state vector in this case is $[P_t, S_t]^T$. Rewrite the system dynamics in matrix form:

$$\begin{bmatrix} P_{t+1} \\ S_{t+1} \end{bmatrix} = \begin{bmatrix} I_2 & \delta t I_2 \\ 0_{2\times 2} & I_2 \end{bmatrix} \begin{bmatrix} P_t \\ S_t \end{bmatrix} + \begin{bmatrix} 0_{2\times 2} \\ \delta t I_2 \end{bmatrix} u_t + \begin{bmatrix} 0_{2\times 1} \\ W_t \end{bmatrix}$$

Measurement model:

$$Y_t = \begin{bmatrix} I_2 & 0_{2\times 2} \end{bmatrix} \begin{bmatrix} P_t \\ S_t \end{bmatrix} + V_t$$

given $\delta t = 1s$, we have system matrices:

$$A = \begin{bmatrix} I_2 & I_2 \\ 0_{2\times 2} & I_2 \end{bmatrix}, B = \begin{bmatrix} 0_{2\times 2} \\ I_2 \end{bmatrix}, C = \begin{bmatrix} I_2 & 0_{2\times 2} \end{bmatrix}$$

(b) A few simulation rollouts over 100 s. Look at solution code for implementation.

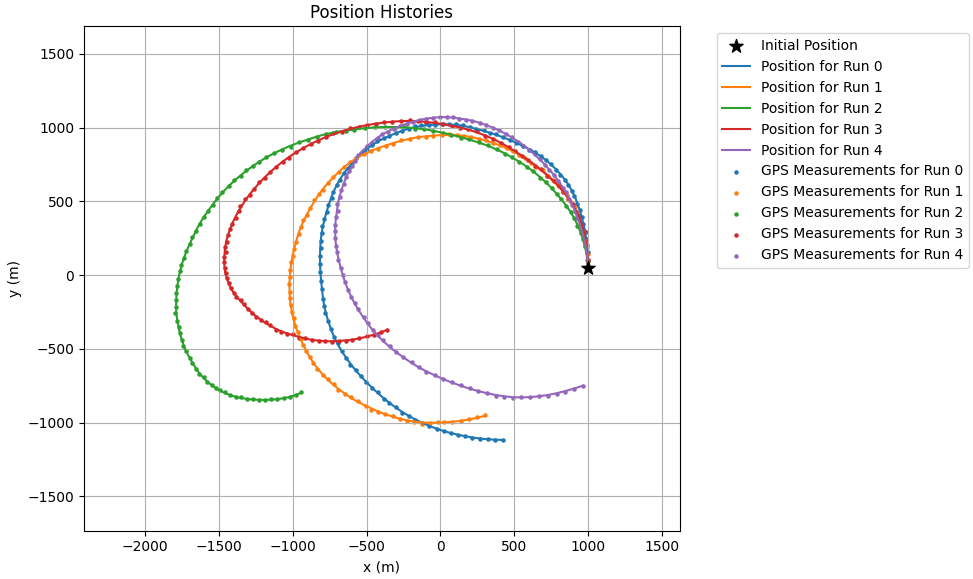


Figure 2: 5 sample trajectories are labeled using different colors

4. [15 Points.] Now you will run a Kalman filter to estimate the drone's position and velocity.

   (a) [5 Points.] Due to the noise only appearing in the velocity equation, we must change one of the Kalman filter equations slightly to accomodate this system. What is the new expression?

   (b) [9 Points.] Suppose your initial guess about the state of the drone has a mean $\mu_0 = [1500m \quad 100m \quad 0m/s \quad 55m/s]^T$, and covariance matrix $\Sigma_0 = [250000\text{I}m^2 \quad 0; \quad 0 \quad \text{I}m^2/s^2]$ (i.e. we are pretty sure about its velocity, but not its position). Use the Kalman filter to find your estimated state trajectory. Use your error ellipse script from PS2 to plot the means and error ellipses of the position estimate over the top of the true trajectory.

   (c) [1 Point.] Similarly, plot the mean and error ellipses of the velocity along the trajectory of the drone (plotting them as vectors emanating from the drone's position, with the ellipse centered at the tip of the vector). We have essentially used the Kalman filter as a very fancy numerical differentiator to find velocity from GPS.

Solution

(a) The given covariance matrix Q corresponds to gaussian random vector W_t which is of dimension 2×1. However, the state vector is 4×1. To accommodate this system, Gaussian noise in dynamic distribution equation should be extended to 4×1. Thus, system equation can be expressed as:

$$\begin{bmatrix} P_{t+1} \\ S_{t+1} \end{bmatrix} = \begin{bmatrix} I_2 & \delta t I_2 \\ 0_{2\times 2} & I_2 \end{bmatrix} \begin{bmatrix} P_t \\ S_t \end{bmatrix} + \begin{bmatrix} 0_{2\times 2} \\ \delta t I_2 \end{bmatrix} u_t + \begin{bmatrix} 0_{2\times 1} \\ W_t \end{bmatrix}$$

Covariance matrix corresponding to the new Gaussian noise can be expressed as:

$$Q_{new} = \begin{bmatrix} 0_{2\times 2} & 0_{2\times 2} \\ 0_{2\times 2} & Q \end{bmatrix}$$

Corresponding expression for predict process:

$$\Sigma_{t+1|t} = A_t \Sigma_{t|t} A^T + Q_{new}$$

(b) See solution code for implementation. An example rollout with position covariances shown as confidence ellipse is shown in Figure 3. Note that the initial covariance and mean error is extremely high, but as soon as we get our first measurement this covariance and mean error reduces significantly!

(c) See solution code for implementation. An example rollout with velocity estimates shown as vectors with error ellipses at their tips is shown in Figure 4

5. [5 Points.] Consider the same model of the drone, but we only have a velocity sensor with the measurement equation

$$Y_t = S_t + V_t.$$

The parameters are the same, and the initial state estimate is $\mu_0 = [1000m \quad 0m \quad 0m/s \quad 50m/s]^T$ with covariance $\Sigma_0 = [\text{I}m^2 \quad 0; \quad 0 \quad \text{I}m^2/s^2]$ (i.e. we are pretty sure about the initial state). Run the Kalman filter to estimate the position and velocity in this case. Plot them in the same way as before. Here we have essentially used the Kalman filter as a very fancy numerical integrator to get position from velocity measurements.[1]

Solution

See solution code for implementation. An example rollout with position covariances shown as confidence ellipse is shown in Figure 5. A very interesting trend here is that, since we are now essentially integrating noisy velocity measurements, our position covariances continue to increase, and our means drift from the true trajectory. An example rollout with velocity estimates shown as vectors with error ellipses at their tips is shown in Figure 6

[1] This is called "dead reckoning," and as you can see, it cannot provide good position estimates for very long.

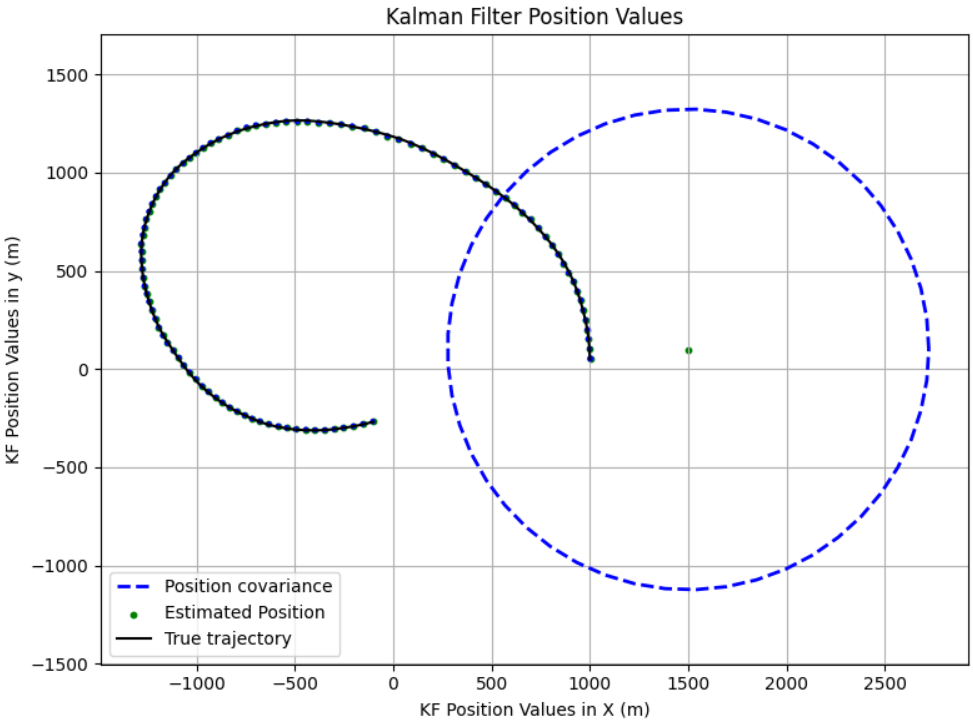


Figure 3: Mean and error ellipse plot for KF position estimates

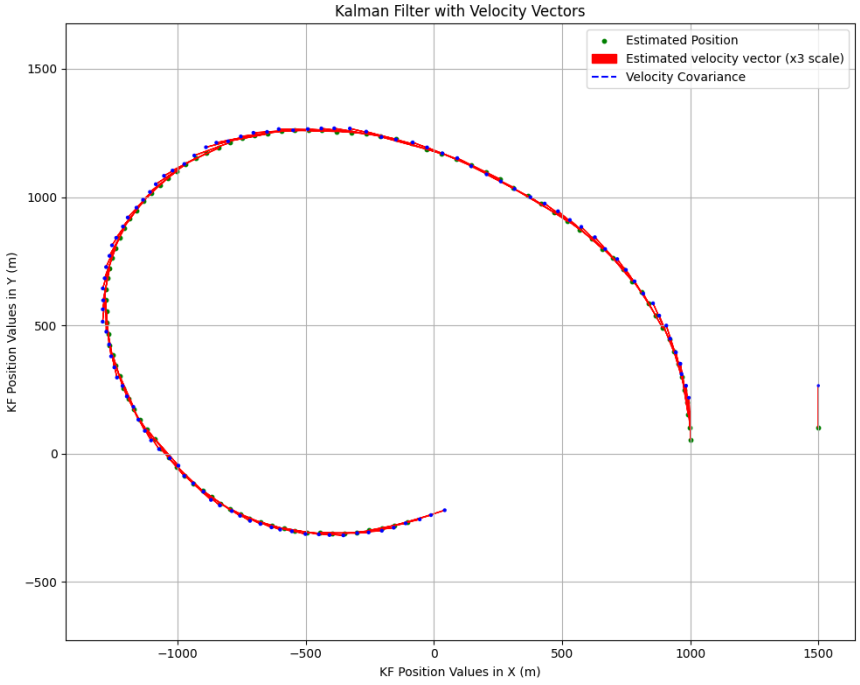


Figure 4: Mean and error ellipse plot for KF velocity estimates, shown over the estimated trajectory.

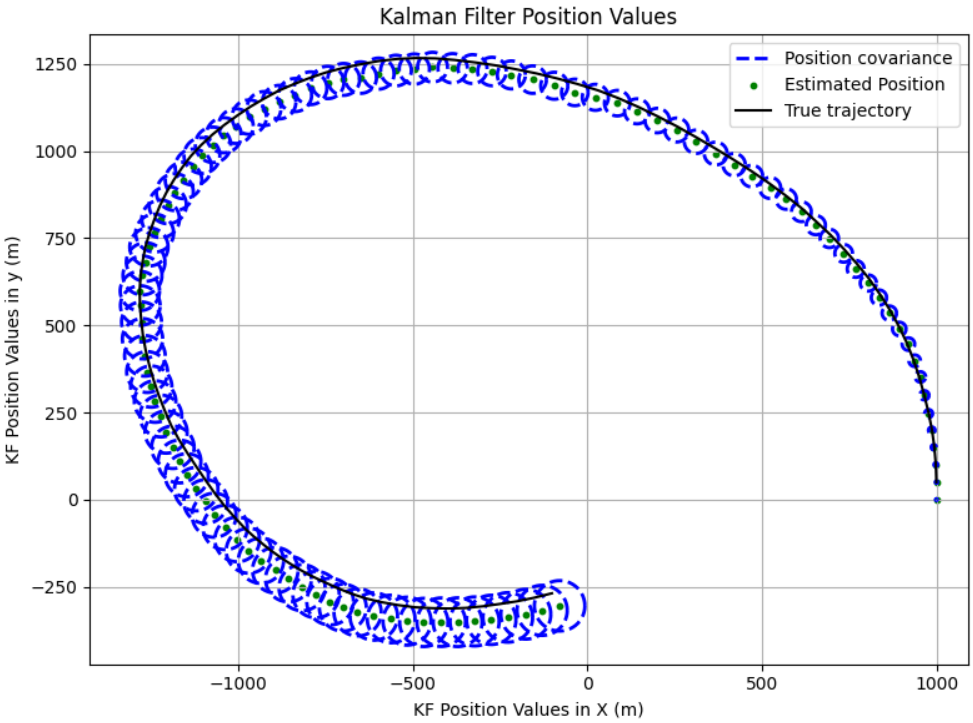


Figure 5: Mean and error ellipse plot for KF position estimates

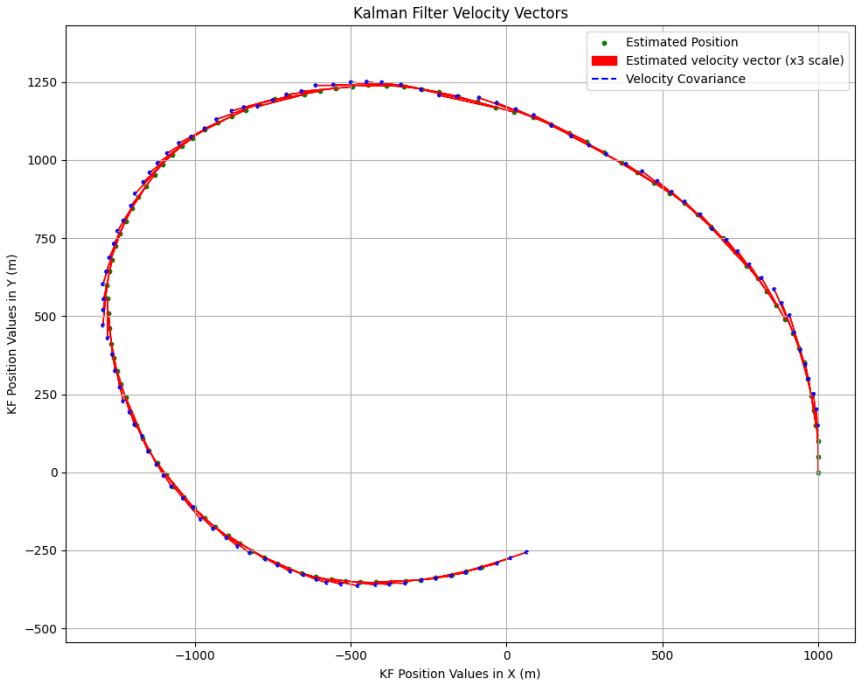


Figure 6: Mean and error ellipse plot for KF velocity estimates, shown over the estimated trajectory.